

Technical Report: k-NN Search in Vector Databases with Multi-Attribute Fairness Constraints

1 TECHNICAL PROOFS

In Section 1.0.1, we present the proofs of the qualitative guarantees of the near candidate retrieval algorithms. In Section 1.0.2, we present the proofs of the qualitative guarantees of the postprocessing algorithms as a stand alone framework. In Section 1.0.3, we evaluate the quality of the retrieved obtained using *Alg-Near-Neighbor*, followed by the postprocessing algorithms. Specifically, we first establish that the proposed postprocessing algorithms are guaranteed to produce solutions that satisfy the defined fairness constraints over the protected attributes, provided that such a feasible fair solution exists in the data. The only source of approximation arises in the computation of distances, which, however, can be analytically bounded.

1.0.1 Qualitative Guarantees of Near Candidate Retrieval Algorithms.

THEOREM 1.1 (NUMBER OF HASH TABLES IN *ALG-NEAR- π*). *Given an (R, cR, p_1, p_2) -sensitive hash family, let n_π denote the number of points in partition π and μ_π the concatenation length, where*

$$\mu_\pi = \frac{\log n_\pi}{\log(1/p_2)} \quad \text{and} \quad \rho = \frac{\log(1/p_1)}{\log(1/p_2)}$$

[1]. Let K be the maximum number of near points to be retrieved for any query with success probability at least $1 - \frac{\delta}{2}$. Then the required number of hash tables for partition π , denoted ℓ_π , must satisfy:

$$\ell_\pi \geq n_\pi^\rho \log(2K/\delta) \quad (1)$$

PROOF. Note that $p_1^{\mu_\pi} = p_1^{\frac{\log n_\pi}{\log(1/p_2)}} = e^{\log p_1 \cdot \frac{\log n_\pi}{\log(1/p_2)}} = n_\pi^{-\rho}$. For a single near point x with $\|x, q\| \leq R$, the probability that x fails to collide with the query in any of the ℓ_π tables is at most $(1 - p_1^{\mu_\pi})^{\ell_\pi} \leq e^{-\ell_\pi p_1^{\mu_\pi}} = e^{-\ell_\pi n_\pi^{-\rho}}$. Applying a union bound over all K true neighbors yields a total failure probability bounded by $Ke^{-\ell_\pi n_\pi^{-\rho}}$. We expect this failure probability is at most $\delta/2$, i.e., $Ke^{-\ell_\pi n_\pi^{-\rho}} \leq \delta/2$. Thus, $\ell_\pi \geq n_\pi^\rho \log(2K/\delta)$. $\square$

Another failure mode arises when the retrieval set is dominated by false positives—points whose distance from the query exceeds cR . Since retrieving exactly k_π points may not guarantee the presence of k_π true near neighbors, it is necessary to examine more than k_π candidates. In this case, we bound the expected number of false positives and determine the minimum number of retrieved points required to ensure that at least k_π near points are included.

THEOREM 1.2 (EXPECTED NUMBER OF FALSE POSITIVES.). *With probability at least $1 - \frac{\delta}{2}$, there are at most $\lceil \frac{2\ell_\pi}{\delta} \rceil$ false positives in the retrieval set. Hence, the minimum number of points to guarantee at least k_π near point is given by:*

$$k_\pi^* = k_\pi + \left\lceil \frac{2\ell_\pi}{\delta} \right\rceil \quad (2)$$

PROOF. Let Y be a random variable denoting the number of pairs (x, j) with $\|x, q\| > cR$ and $g_j(x) = g_j(q)$. Since each false positive happens with probability p_2^μ , we have:

$$\mathbb{E}[Y] \leq n_\pi \ell_\pi p_2^\mu = n_\pi \ell_\pi \frac{1}{n_\pi} = \ell_\pi \quad (3)$$

By Markov inequality:

$$\mathbb{P}[Y > t \cdot \ell_\pi] \leq \frac{1}{t} \quad (4)$$

We expect $\frac{1}{t} = \frac{\delta}{2} \Rightarrow t = \frac{2}{\delta}$. The inequality becomes:

$$\mathbb{P}\left[Y > \frac{2}{\delta} \cdot \ell_\pi\right] \leq \frac{\delta}{2} \quad (5)$$

Or, equivalently:

$$\mathbb{P}\left[Y \leq \frac{2\ell_\pi}{\delta}\right] \geq 1 - \frac{\delta}{2} \quad (6)$$

Thus, retrieving $k_\pi^* = k_\pi + \lceil \frac{2\ell_\pi}{\delta} \rceil$ nearest neighbors from the partition π guarantees the appearance of at least k_π near points, where ℓ_π is given by Theorem 1.1. $\square$

THEOREM 1.3 (RUNNING TIME OF *ALG-NEAR- π*). *Let partition π have n_π vectors, each of d -dimensional, and ℓ_π LSH tables using μ concatenated base hashes per table. Let k_π be the quota for π , and let k_π^* be defined by Eq. 2. Assume ℓ_π satisfies Eq. 1, with probability at least $1 - \delta$, *Alg-PostP- π* runs in $O(\ell_\pi \mu d + k_\pi^* d + k_\pi^* \log k_\pi)$.*

PROOF. A compound hash function hashes a query μ times across ℓ_π hash tables, each hash operation is $O(d)$. Hence, the total cost of hashing query is $O(\ell_\pi \mu d)$. After probing k_π^* candidates, we compute the true distance and perform sorting to find top- k_π , which take $O(k_\pi^* d)$ and $O(k_\pi^* \log k_\pi)$ time, respectively. $\square$

1.0.2 Qualitative Guarantees of the Postprocessing Algorithms. We prove below that the proposed postprocessing algorithms exactly satisfy the fairness constraints, provided a feasible fair solution of size k exists in the candidate set C .

THEOREM 1.4. *Post-*Alg-1* is optimal given the candidate set C .*

PROOF. Fairness is imposed on a single protected attribute A_j with domain V_j and ranged constraints $\{[l_{j,v}, u_{j,v}]\}_{v \in V_j}$. The candidate set C is partitioned into disjoint subsets

$$C_v = \{x_i \in C : a_{i,j} = v\}, \quad v \in V_j.$$

*Post-*Alg-1** retains the $u_{j,v}$ closest points from each C_v as the per-group pool P_v , mandatorily selects the $l_{j,v}$ closest points from P_v , and fills the remaining budget $r = k - \sum_v l_{j,v}$ greedily from the residual pool $\bigcup_v \{\text{unselected items of } P_v\}$ sorted globally by distance. Thus the final solution S satisfies $|S| = k$ and, for each $v \in V_j$,

$$l_{j,v} \leq |\{x_i \in S : a_{i,j} = v\}| \leq u_{j,v}.$$

Hence all ranged fairness constraints on A_j are satisfied whenever $|C_v| \geq l_{j,v}$ for all v , i.e., whenever a feasible fair solution exists in C . Optimality follows because the mandatory phase selects the closest

$l_{j,v}$ items per group, and the greedy phase selects the globally closest items from the residual pool subject to per-group caps — any other feasible selection would have weakly larger total distance. $\square$

THEOREM 1.5. *Post-Alg-2 is optimal given the candidate set C .*

PROOF. Because all demands and capacities are integral, any feasible min-cost flow is integral. Consider an integral Source-Sink flow f of value k . Flow conservation and the capacity bounds on the relay edges imply:

$$l_{i,v} \leq \sum_{p: a_{p,i}=v} f_p \leq u_{i,v}, \quad l_{j,t} \leq \sum_{p: a_{p,j}=t} f_p \leq u_{j,t}$$

We define the selected set $S_f = \{x_p \in C : f_p = 1\}$. Then:

$$|S_f| = k, \quad l_{i,v} \leq |\{x_p \in S_f : a_{p,i} = v\}| \leq u_{i,v}, \quad l_{j,t} \leq |\{x_p \in S_f : a_{p,j} = t\}| \leq u_{j,t}$$

Thus, S_f satisfies all ranged fairness constraints.

To show that S_f optimizes the total distance to the query, note that any set of points that satisfy the fairness constraints defines a feasible flow (namely, a flow that satisfies the constraints in Eq. ??) whose cost is the total distance to the query. Since S_f achieves minimum cost, it optimizes the total distance to the query. $\square$

THEOREM 1.6. *Post-Alg-3+ is optimal given the candidate set C .*

PROOF. (sketch.) It could be shown that a feasible solution of the ILP satisfies all ranged fairness constraints, i.e., $l_{j,v} \leq \sum_{i: a_{i,j}=v} z_i \leq u_{j,v}$ for all (j, v) . If a feasible fair k -set exists in C , the ILP has a feasible solution, and the ILP solution returned is optimal. $\square$

1.0.3 Qualitative Guarantees of the Overall Algorithmic Framework.

THEOREM 1.7. *If all near points are retrieved with probability at least $1 - \frac{\delta}{2}$ from a partition using Alg-Near-Neighbor and then postprocessed using the algorithms in Section ??, then the probability that Alg-1-Fair Alg-2-Fair Alg-3+-Fair return C approximate nearest neighbors of a query is $1 - \frac{\Pi\delta}{2}$.*

PROOF. Suppose that for each partition π , all near points are retrieved with probability at least

$$1 - \frac{\delta}{2}.$$

Let there be Π partitions in total for a query. For any fixed partition π , the failure probability is therefore

$$\Pr[\text{fail on } \pi] \leq \frac{\delta}{2}.$$

By the union bound, the probability that at least one partition fails is

$$\Pr[\exists \pi \text{ that fails}] \leq \sum_{\pi=1}^{\Pi} \Pr[\text{fail on } \pi] \leq \Pi \cdot \frac{\delta}{2}.$$

Hence, the probability that *all* partitions successfully retrieve all near points is

$$\Pr[\text{all partitions succeed}] = 1 - \Pr[\exists \pi \text{ that fails}] \geq 1 - \frac{\Pi\delta}{2}.$$

$\square$

REFERENCES

- [1] Mayur Datar, Nicole Immorlica, Piotr Indyk, and Vahab S Mirrokni. 2004. Locality-sensitive hashing scheme based on p-stable distributions. In *Proceedings of the twentieth annual symposium on Computational geometry*. 253–262.